H_2O_2 Low Temperature Plasma Sterilizer

Operation Manual

PS-100X/200X

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Welcome use H_2O_2 Low Temperature Plasma Sterilizer. To maximize the performance of the equipment, please carefully read and comply with this manual for safety and operational purpose.

This manual is applicable to PS-100X / 200X H_2O_2 Low Temperature Plasma Sterilizer.

Attention:

Keep this manual perfectly during the service life of the equipment.

Make sure all updated information received has been included in this manual.

This manual must be transferred or handed over as part of the equipment whenever the equipment occupancy site is altered.

Product description

H_2O_2 Low Temperature Plasma Sterilizer controlled by micro-computer program automatically. It consists of sterilizing chamber, vacuum pump, H_2O_2 injection system and high-frequency power supply. 56%-63% of H_2O_2 is adopted as sterilizing agent to kill bacterial spore. The sterilizer could perform sterilization in a short period under the condition of low temperature and humidity.

Sterilization principle

The sterilizer adopts H_2O_2 as sterilizing agent and form plasmatic state of H_2O_2 by electromagnetic field under low temperature. It combines both gaseous and plasmatic H_2O_2 to sterilize the items in chamber and decomposes the residual H_2O_2 after that.

Category of killed microorganism

Bacterial spore.

Major factor of sterilization

56%-63% H_2O_2.

Scope of application

The sterilizer is applicable to sterilize the medical instruments that are thermolabile or non-moisturetight. It is not applicable to the sterilization of liquid, powder or cotton fabrics. All the medical instruments that meet the following conditions can be processed by this sterilizer.

PS-100X:

Metal tubular instrument: inner diameter≥1mm; length≤500mm

Nonmetal tubular instrument: inner diameter≥1mm; length≤2000mm

PS-200X:

Metal tubular instrument: inner diameter≥0.8mm; length≤300mm

Nonmetal tubular instrument: inner diameter≥1mm; length≤1000mm

Technical parameters

PS-100X

Model	Working power supply	Range of working temp.	Sterilization cycle	Service life
PS-100X	AC 380v,50Hz	＜60℃	50 min—65min	6 years
Power	Rated current	Chamber size	Overall size	Weight
3Kw	13A	(75×45×40)cm	(99×76×160)cm	400Kg

PS-200X

Model	Working power supply	Range of working temp.	Sterilization cycle	Service life
PS-200X	AC 380v,50Hz	＜60℃	50 min—65min	6 years

Power	Rated current	Chamber size	Overall size	Weight
3.5Kw	15A	(125×45×40)cm	(146×76×160)cm	500Kg

Usage

Please strictly obey the instruction that mentioned in this manual to operate the sterilizer.

1) Turn on the sterilizer and the touch-screen works;

2) Click the Open button to open the chamber. Place the items into chamber and close it. Back to the main menu;

3) Click the 'Program' button;

4) Click the Start button to start the sterilizer;

5) The sterilization stops after a full cycle program;

6) When the chamber pressure returns to the local atmospheric pressure the buzzer sounds while sterilization finished.

Sterilization procedure of rapid cycle

Vacuum→ pressure holding→ H_2O_2 injection→ purification→ transmit→ diffusion→ vacuum→ plasma→ vacuum breaking

Notice

1) We do not recommend sterilize heat-resistant or moisturetight items by this low temperature sterilizer but pressure steam sterilizer. The following items are not applicable to this sterilizer: hygroscopic materials (i.e. wooden instrument, cellulose, cotton fabric, gauze, etc.); disposable instrument (i.e. disposable waterproof fabric, disposable surgical gown, etc.); oxidizable items (i.e. silk, paper articles, etc.); liquid or powder; sealed tubular instrument.

2) Preparation work before sterilization:

- Rinse the instruments by pure water and thoroughly dry them.
- Pack the instruments by single-layer paper-plastic bag or double-layer nonwovens.

3) Place the packed instruments without overlap. Leave appropriate gap between packs.

4) The packs that placed in the chamber should not exceed 80% of the chamber capacity.

5) Please pay attention the fault prompt during the sterilization. If any, debug it and restart.

6) Do not pull the chamber door forcibly after sterilization until the buzzer sounds.

7) The H_2O_2 that contained in the sterilizing agent could easily permeate into skin. Please wear gloves that are made of latex or ethylene while handle the sterilizing agent. Handle the used cassettes properly.

8) The standard H_2O_2 content of the sterilizing agent is 56%-63%. Please test the H_2O_2 content by concentration indicator or specified testing method. The sterilizing agent that used in this sterilizer must be permitted by local health institutions.

9) Before sterilization, please make sure the H_2O_2 cassette is within its validity.

Conditions of transport and storage

Temperature: 10℃~40℃

Relative humidity: ≤80%

Atmospheric pressure: 70kPa~106kPa

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1 Cautions and attentions

Please strictly obey the following instructions during the operation.

1.1 Attentions

Read carefully the following information while operating or maintaining the sterilizer. Please pay attention to these signs wherever they appear.

It indicates that please pay high attention.

It indicates that please obey or damage may happens on sterilizer.

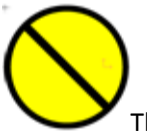
It indicates that please strictly obey or human safety may be endangered.

Operators should always keep the following information in mind before operation.

Wherever the sign appears, please look up the relevant information in this manual to clarify the potential danger and take proper measures.

Anyone in any situation is not allowed to tear off any page of this manual. Please contact us promptly if this manual cannot help you with the sterilizer problems. Please make sure your manual is updated.

Please store this manual at ventilated and dry place preventing damps and heat.


The sterilizing agent contains H2O2 which could permeate into and temporarily skin. Please wear gloves that are made of latex or ethylene while handle the sterilizing agent. Handle the used cassettes properly. In any case of that your skin contacts the sterilizing agent; please wash it by plenty of water immediately for 5 minutes. If anyone accidently swallows the agent please drink plenty of water and turn to the doctor.

The following articles (including but not limited) mentioned are not applicable to this H2O2 low temperature sterilizer:

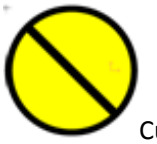
a. **PS-100X**:

The stainless steel tubular instruments with inner diameter＜1mm and length＞500mm.
The Teflon tubular instruments with inner diameter＜1mm; length＞2000mm
PS-200X:
The stainless steel tubular instruments with inner diameter＜0.8mm and length＞300mm.
The Teflon tubular instruments with inner diameter＜1mm; length＞1000mm
b. Hygroscopic materials (i.e. wooden instrument, cellulose, cotton fabric, gauze, etc.);
c. Disposable instrument (i.e. disposable waterproof fabric, disposable surgical gown, etc.);
d. Liquid, cream, oil, and powder;
e. Half-dried items;
f. The tubular instruments with one sealed end;
g. Implants;
h. The instruments that could not bear vacuum pressure;
i. The instruments that are only allowed to be sterilized by pressure steam sterilizer or EO sterilizer;
j. The instruments that contain complicated construction e.g. sealed bearing;
k. The instruments that contain nylon surface;
l. The soft endoscopes with fossilized or damaged surface that their materials cannot be determined.
We do not make any explicit or implied guarantees or warranties for any damage caused by biogenetic or secondary injury and improper operation.
Please inquire our engineering department if you are not sure whether your instrument can be processed by this sterilizer.

All instruments must be thoroughly washed and dried, and properly packed by specified packing material or container before being put into chamber for sterilization.

To guarantee the performance of the sterilizer, it is necessary to carry out regular and preventive maintenance.

Any product adjustment or maintenance must be left to qualified staff. Improper maintenance or unauthorized installation of spare parts may cause personal injury.


Cut off all power supply of the sterilizer before maintenance.

1.2 Safety signs

On (power on) IEC 417 , No. 5007	Off (power off) IEC 417, No. 5008	Ground IEC 417 , No. 5017	Protective ground IEC 417, No. 5019
DC IEC417 , No. 5031	AC IEC417 , No.5032	Refer to the attached materials ISO3864 No.B.3.1	Prevent electric shock ISO3864, No. B.3.6
Prevent surface overheat IEC 417 , No. 5041	Refer to user manual YY0466 , No. 3.3	Keep dry YY0466 , No. 3.8	Temperature limitation YY0466 ,No. 3.11
Avoid direct sunlight YY0466,No. 3.6	Avoid high voltage IEC417 , No.5036	Fuse IEC 417, No. 5016	Keep in upright position GB/T191 ,No. 3

2 Description of H2O2

2.1 Nature of H2O2 solution

SHINVA low temperature plasma sterilizer adopts 56%~63% H_2O_2 solution to kill bacterial spore. H_2O_2 is a colorless and ropy liquid which has strong oxidability. The H_2O_2 solution is faintly acid. H_2O_2 solution is an explosive strong oxidizer. It is incombustible whereas can react with combustibles to release plenty of heat and gas so that cause fire and explosion.

2.2 Storage of H2O2 cassette

The H2O2 cassette should be stored in ventilated and dry place with the room temperature of 25℃. It must be prevented from direct sunlight and crash and pressing. Store the H2O2 cassette away from combustibles, heavy metal, metal compounds, heat source and combustion source. The cassette must be used within the period of validity to guarantee the sterilization performance.

The used cassette must be properly disposed according to local regulation.

2.3 Emergency measures

The H_2O_2 has strong oxidability that could easily permeate into skin and cause temporary injury.

Please wear gloves that are made of latex or ethylene while handle the sterilizing agent.
In any case of that your skin contacts the sterilizing agent; please wash it by plenty of water immediately for 5 minutes. If anyone accidently swallows the agent please drink plenty of water and turn to the doctor.

To change the cassette or handle sterilized items please wear gloves that are made of latex or ethylene. Do not touch your face or eyes while wearing the gloves.

3 Introduction of H2O2 plasma sterilization

3.1 Sterilization principle

H_2O_2 plasma sterilization is a new technology of low temperature sterilization in recent science of disinfection. With the development of medical and biological high-tech, the existing sterilization technique cannot meet some particular requirements. Some of precise medical instrument (i.e. fiber scope) or thermolabile material is only applicable to low temperature sterilization technique. Comparing with HCHO, EO and glutaraldehyde that have the disadvantages of low efficiency and high toxicity, the relatively efficient and nontoxic H_2O_2 plasma sterilization becomes preference in disinfection industry.

The process of H_2O_2 low temperature plasma sterilization: the vacuum degree in chamber reaches a certain value→ turn on the valve which in between the H_2O_2 cassette and the chamber→ the liquid H_2O_2 diffuses in chamber after heated and purified→ bacteria and virus will be killed during the H_2O_2 diffusion→ under the effect of high frequent electric, gaseous H_2O_2 is dissociated into charged particle so that turns into plasma→a large number of ultraviolet light is generated during the formation of plasma to destroy the microorganism→the photolysis of ultraviolet light breaks the chemical bond of microorganism and generate volatile compounds (i.e. CO, CHx)→destroy the vital force of microorganism by plasma effect.
In this process, the temperature of electron is high while that of ion is almost as low as room temperature. This phenomenon has fulfilled low temperature sterilization instead of sterilizing items by high-temp sterilization.

3.2 Sterilization advantages

H2O2 low temperature sterilizer adopts H2O2 agent and plasma to perform sterilization under low temperature (about 50℃) and pressure. It is perfectly applicable to the sterilization of thermolabile or non-moisturetight medical instrument.
The process of H_2O_2 sterilization is environment friendly with no contamination which has only discharge small amount of oxygen and vapor. There is no need of additional air interchanger or ventilator.
Comparing with HCHO or EO sterilization, H_2O_2 sterilization has relatively short working period (about only 1 hour). It is favorable for emergent sterilization and advances the instrument usage turnover rate.

3.3 Application scope of sterilization

Before sterilization, please make sure that the items that to be sterilized are applicable

to H2O2 low temperature sterilization.

1) Acceptable materials:

Aluminum, copper, Kraton copolymer, glass, ethylene, vinyl acetate, liquid crystal copolymer, polycarbonate, Ti, polyurethane, PVC, stainless steel, PEEK, teflon, PEI (U1tem copolymer), polystyrene, polypropylene, PMMA and siloxane elastomer.

2) Acceptable instruments (including but not limited):

Wires that used to connected to patient, optical glasses, glass lens, rigid endoscope, catheter, surgical instrument, diagnosis instrument.

3) Unacceptable instruments (including but not limited):

a. **PS-100X:**

The stainless steel tubular instruments with inner diameter＜1mm and length＞500mm.

The Teflon tubular instruments with inner diameter＜1mm; length＞2000mm

PS-200X:

The stainless steel tubular instruments with inner diameter＜0.8mm and length＞300mm.

The Teflon tubular instruments with inner diameter＜1mm; length＞1000mm

b. Hygroscopic materials (i.e. wooden instrument, cellulose, cotton fabric, gauze, etc.);

c. Disposable instrument (i.e. disposable waterproof fabric, disposable surgical gown, etc.);

d. Liquid, cream, oil, and powder;

e. Half-dried items;

f. The tubular instruments with one sealed end;

g. Implants;

h. The instruments that could not bear vacuum pressure;

i. The instruments that are only allowed to be sterilized by pressure steam sterilizer or EO sterilizer;

j. The instruments that contain complicated construction e.g. sealed bearing;

k. The instruments that contain nylon surface;

l. The soft endoscopes with fossilized or damaged surface that their materials cannot be determined.

3.4 Instrument treatment before sterilization

3.4.1 Rinse

Proper rinse gives good sterilization performance.

Rinse is a key factor of sterilization especially for some instruments that difficult to clean. According to recent research, thorough rinse may reduce 3~4 logarithms of bacteria amount so that dramatically decrease the content of organics. The activity of sterilizing agent will be highly weakened if the organics have not been washed off. It may result in a poor or unsuccessful sterilization.

- Clean the instrument

Wash the instrument according to the instruction given by the instrument manufacturer. Please apply proper washing detergent.

- Thorough rinse

Totally rinse the instrument to get rid the residual detergent off.

The residual detergent on instrument may leave stain.

3.4.2 Dry

The instrument must be thoroughly dried before sterilization; otherwise, it may cause a delay or interrupt of sterilization.

To dry the instrument, the acceptable method is that blow compressed air into the instrument till there is no water drips out. Besides, you can also dry the instrument by following the method instructed by the instrument manufacturer.

3.4.3 Packing

Do not pack the instrument by paper bag or any fiber or cotton material.

- Acceptable packing material

The paper-plastic packing bag specially designed for H_2O_2 low temperature plasma sterilization, polypropylene nonwovens.

- Metal sterilizing container

We only recommend SHINVA sterilizing container and relevant accessories. The utility of this container is to guarantee the H_2O_2 and plasma could effectively diffuse among the instruments in chamber.

3.4.4 Loading

- The instruments loaded are not allowed to exceed 80% of the chamber capacity. There should not be overlaps between packs but a 10mm gap.
- The paper-plastic bags, if laid in single layer, make sure the paper side up and no overlaps between bags. If they aslant laid in 45°, make sure the paper side up. The paper side should be leaned against the other paper side; similarly, the plastic side must be opposite to the other plastic side.
- The items in sterilizing container must be laid in single layer without overlap. The container should be packed by nonwovens in double-layer and laid in single layer with no overlap.

Any items in chamber must be kept away from chamber wall, chamber door or electrode mesh.

3.5 Detection of sterilization performance

3.5.1 Physical detection

According to relevant detection standards of sterilization performance, each critical parameter of sterilization cycle, such as chamber pressure, temperature and time, should be monitored and recorded. All the sterilizing parameters should accord with the value given by the user manual.

The build-in printer could help users record and print sterilizing data for filing purpose. The printed paper indicates relevant sterilizing parameters (i.e. pressure, temperature, time, etc.) to help to judge the sterilization performance in preliminary stage.

The physical detection is not sufficient to factually reflect the sterilization performance of each

package in chamber. Please detect it combine with chemical and biological methods.

3.5.2 Chemical detection

According to relevant detection standards of sterilization performance, chemical indicator should be applied as a mark of sterilization on each package. Put the indicator in the point of most difficult to sterilize of the package and judge the sterilization performance by observing its color change.

Chemical detection is a easier and low-cost approach to test and judge sterilization performance by visually observing the change of indicator status and color. It can be used to find out sterilization failure such as incorrect packing or loading and disfunction of the sterilizer.

3.5.3 Biological detection

According to relevant detection standards of sterilization performance, the biological detection should be carried out at least once a day.

For the usage instruction of the bio-indicator, please refer to the bio-indicator manual.

For the procedures of the biological detection for new installed sterilizer, please refer to the installation manual.

The chemical indicator (label) and bio-indicator must be used with their validity period!

The bio-indicator of plasma sterilization is not allowed to be replaced by that of steam sterilization.

4 Sterilizer introduction

4.1 Environment conditions

4.1.1 Environment

Temperature: +10℃~+40℃

Relative humidity: 30%~75%

Atmospheric pressure: 70kPa~106kPa

4.1.2 Energy

Power supply: acceptable ground wire

PS-100X: 380V/50HZ/13A or 380V/60HZ/13A

PS-200X: 380V/50HZ/15A or 380V/60HZ/15A

4.1.3 Installation space

Floor flatness:＜5mm/m

Distance between both sides of sterilizer and wall: 100mm

Distance between back side of sterilizer and wall: 100mm

Distance between operating side of sterilizer and obstacle: ＞1m

Sterilizer maintenance space: both sides＞1m

4.2 Technical parameters

Order code / Parameters	PS-100X	PS-200X
Chamber size (LxWxH)mm	750x450x400	1250x450x400
Volume (L)	135	225
Overall size (LxWxH)mm	990x760x1600	1470x760x1600

Weight (kg)	400	500
Power supply	380V 50/60Hz	380V 50/60Hz
Power rate	3KW	3.5KW

4.3 Structural feature

4.3.1 Overall structural

The sterilizer consists of following parts:

a) Chamber (includes outer chamber, reticular electrode, basket and heater);

b) Vacuum system (includes vacuum pump, pressure transmitter, solenoid valve and controller);

c) H_2O_2 injection system (includes cassette driving system, cassette puncture system and purifier);

d) Auto controlling parts (includes electric cabinet and touching-screen);

e) Plasma generating device (includes plasma power supply and wires);

f) Chemical indicator, chemical label, packing bag, nonwovens and 60% H_2O_2 cassette for plasma sterilization.

4.3.2 Material advantages

Due to the corrosivity of H_2O_2, the material adopted in this sterilizer has the following advantages:

- Corrosion resistance,
- Guarantee of H_2O_2 quality,
- No contamination or poison generated.

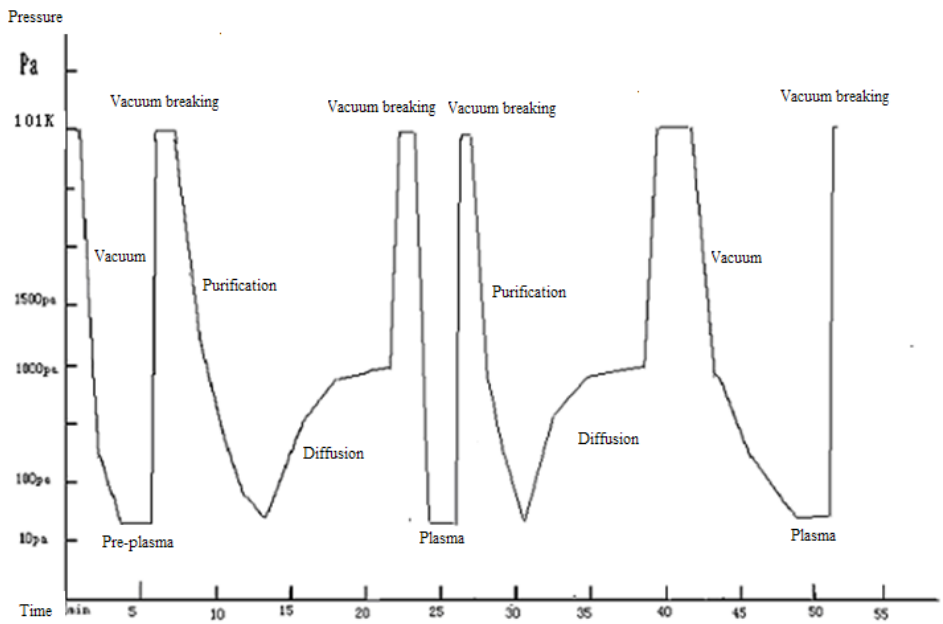
4.3.3 Sterilization procedure

1) Vacuum phase I

Activate the vacuum pump→ chamber pressure reaches certain value→ the phase of pressure holding/pre-plasma starts→ dehumidification and ready to sterilization→ the pressure will be kept in certain rang during set time→ the high-frequency radio source generate plasma.

2) Purification phase I (not applicable for rapid cycle)

Purified air enters into chamber→ the pressure in chamber back to atmospheric pressure→ vacuuming starts while H_2O_2 capsule punctured→ H_2O_2 absorbed into purifier to be purified.



3) Diffusion phase I (not applicable for rapid cycle)

H_2O_2 diffuses in chamber→ pressure in chamber climbs to set range and will be kept till the end of this phase.

4) Vacuum phase II (not applicable for rapid cycle)

Air exchanging (i.e. purified air in)→ pressure in chamber back to atmospheric pressure→ vacuuming starts till the pressure lower than the set value.

5) Plasma phase I (not applicable for rapid cycle)

The pressure in chamber will be kept in certain rang during set time→ the high-frequency radio source generate plasma.

6) Purification phase II

Purification phase I repeated.

7) Diffusion phase II

Diffusion phase I repeated.

8) Vacuum phase III

Vacuum phase II repeated.

9) Plasma phase II

Plasma phase I repeated.

10) Vacuum breaking phase

Purified air enters into chamber→ pressure in chamber back to atmospheric pressure→ end of sterilization.

11) Data printing

- As soon as the sterilization finishes, the buzzer sounds for 4 seconds and the printer starts to print the data.
- The next sterilization cycle starts as soon as the data printing finishes.

4.4 Sterilization cycles

A. PS-100X

1) Full cycle

The instruments that meet the following conditions could be processed by PS-100X sterilizer. The maximum quantity of the instruments in following sizes should not exceed 10.

- The stainless steel tubular instruments with inner diameter ≥1mm and length≤500mm.
- The Teflon tubular instruments with inner diameter≥1mm; length≤2000mm

2) Rapid cycle

This cycle is only applicable to surface sterilization. The instrument to be processed should be solid. The instruments loaded in chamber should not exceed 50% of the chamber capacity.

The following instruments (including but not limit to) are acceptable to be processed by this sterilizer: solid endoscope, rechargeable batteries, and solid ophthalmic instrument.

Material (including but not limit to) that is applicable to full sterilization but rapid cycle: PU, Kraton copolymer, Ultem copolymer and Delrin.

B. PS-200X

1) Full cycle

The instruments that meet the following conditions could be processed by PS-200X sterilizer. The maximum quantity of the instruments in following sizes should not exceed 10.

- The stainless steel tubular instruments with inner diameter≥0.8mm and length≤300mm.
- The Teflon tubular instruments with inner diameter≥1mm; length≤1000mm

2) Rapid cycle

This cycle is only applicable to surface sterilization. The instrument to be processed should be solid. The instruments loaded in chamber should not exceed 50% of the chamber capacity.

The following instruments (including but not limit to) are acceptable to be processed by this sterilizer: solid endoscope, rechargeable batteries, and solid ophthalmic instrument.

Material (including but not limit to) that is applicable to full sterilization but rapid cycle: PU, Kraton copolymer, Ultem copolymer and Delrin.

5 Operation instructions

5.1 Switching system

The switching system of this H_2O_2 low temperature plasma sterilizer consists of power cable, air switch (see P1) and switch button. Preheat the sterilizer in 30 minutes before sterilization every day. The program can be started once the displayed temperature exceeds 45℃. The air switch should be in normal close position. Please cut off the air switch and disconnect the power cable when the sterilizer will not be operated for long time, or to be maintained.

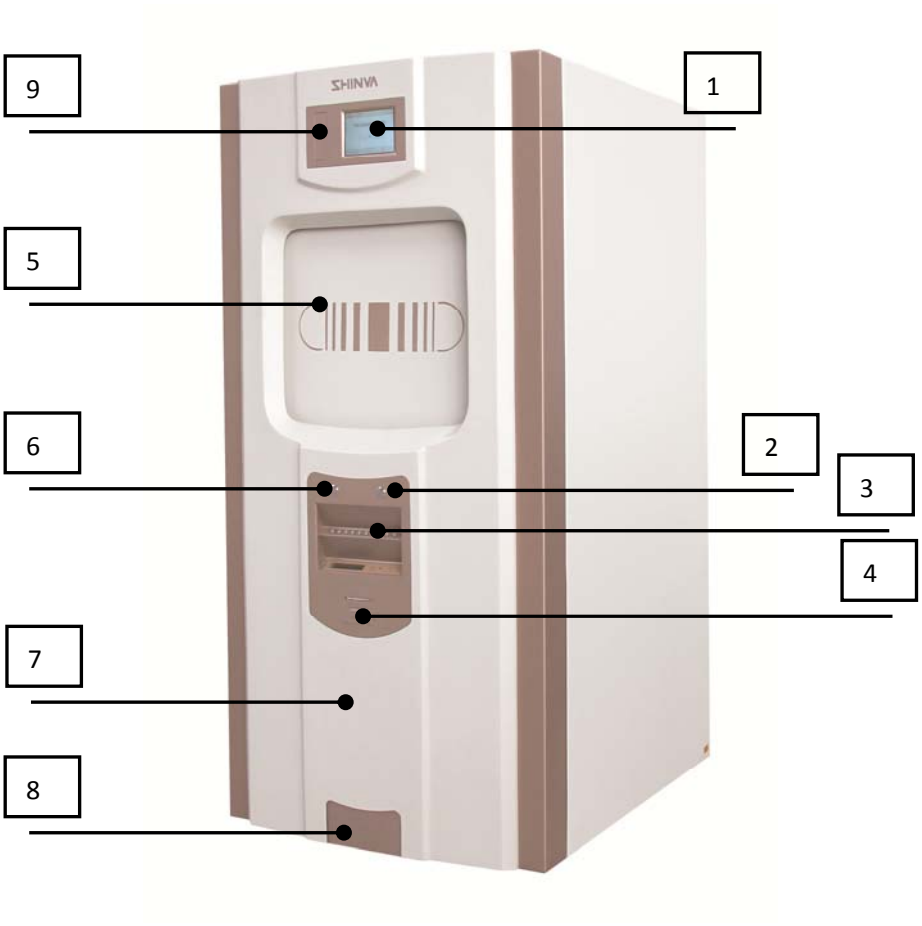
Pic 1 Back view

5.2 Instructions of front/back operation end

Front operation end consists of touch-screen, start button (ON), shut-down button (OFF), door switch button (optional), biological incubator (optional) and foot switch (optional).

Back operation end (only applicable for double-door sterilizer) consists of open-door button,

close-door button, front door indicating light, back door indicating light, operation indicating light, end-of-run indicating light and alarm light.



Pic 2 Front view

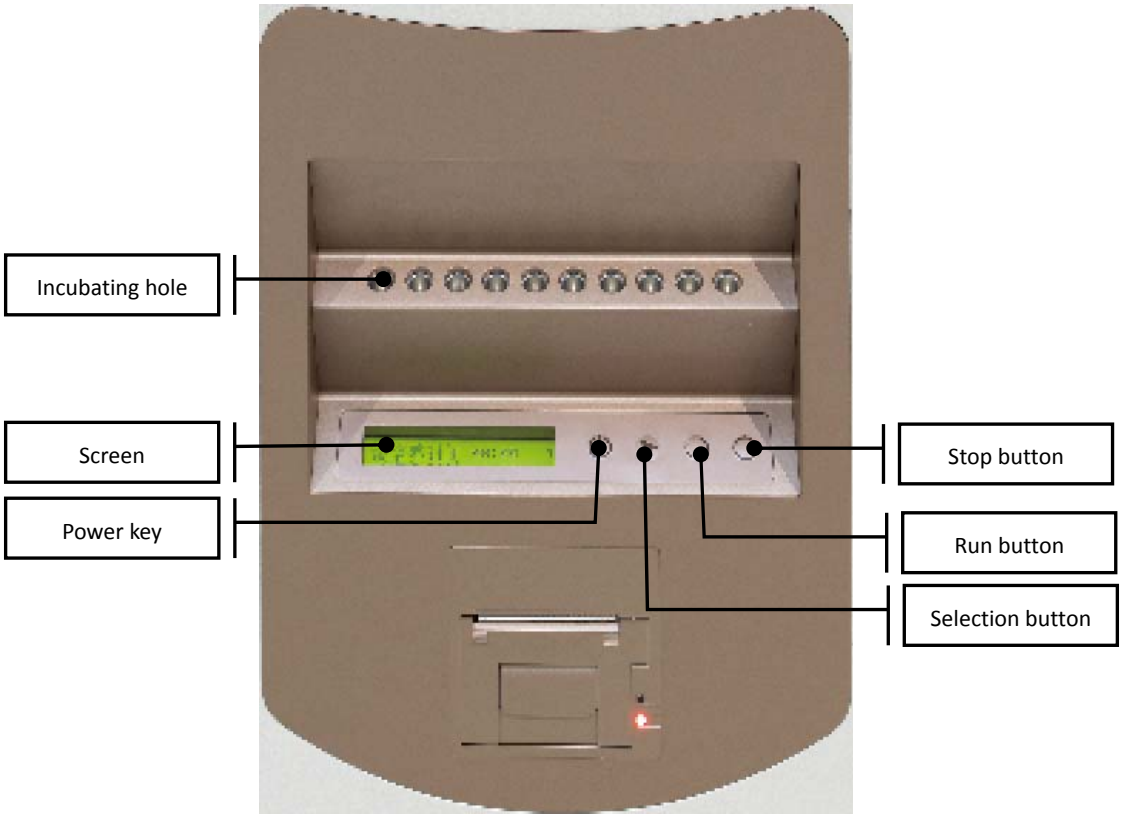
No	Name	Description
1	Touch-screen	Display program process and sterilizer status; program selection.
2	ON/OFF button	See detail a)
3	Biological incubator (optional)	See detail b)
4	Printer	Print sterilizing process and result.
5	Chamber door	Seal the sterilizing chamber.
6	Door button(optional)	Open/close the chamber door
7	Front cover	Decorative cover
8	Foot switch(optional)	See detail c)
9	Cassette slot	Insert or withdraw the cassette.

a) ON/OFF button

Under the condition of that the air switch is closed and the light of OFF button is on, push the ON button to activate the touch-screen.

b) Biological incubator

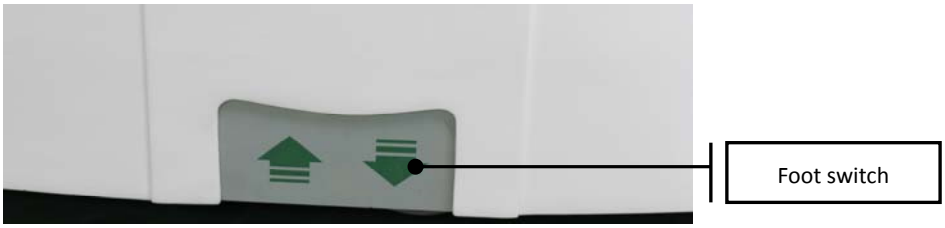
The incubator is designed for incubating biological indicator. The 10 incubating holes are divided into 5 pairs by the control system. This incubator can process 5 pairs of incubation at the same time. Push the power button to heat the incubator till the displayed temperature is 56℃. Select a pair of circulating holes by pushing the selection button (next to the power button) and push the RUN button. Now the screen starts to time the incubation time. Push the STOP button when the incubation is terminated. Push the power button to stop heating the incubator. The incubating result can be stored and printed.



Biological incubator

c) Foot switch

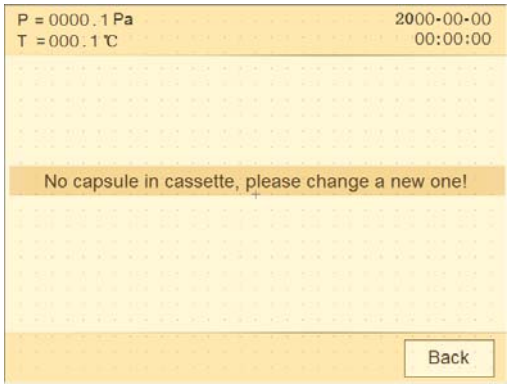
The chamber door can be opened or closed by kicking the foot switch.

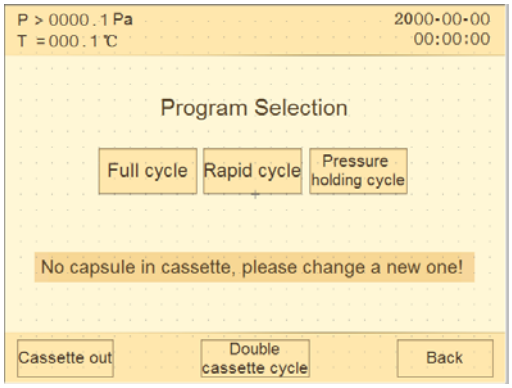


5.3 Insert/withdraw the cassette

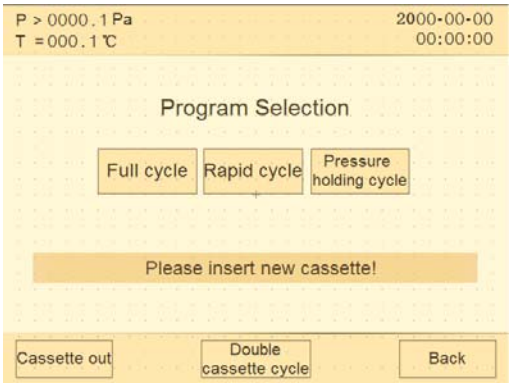
5.3.1 Insert/change cassette

1) The cassette will automatically move out from the slot once the 12 capsules are used out. The screen prompts '**No capsule in cassette, please change a new one!**' Please insert a new cassette or push the 'Return' button to go back to the main menu.

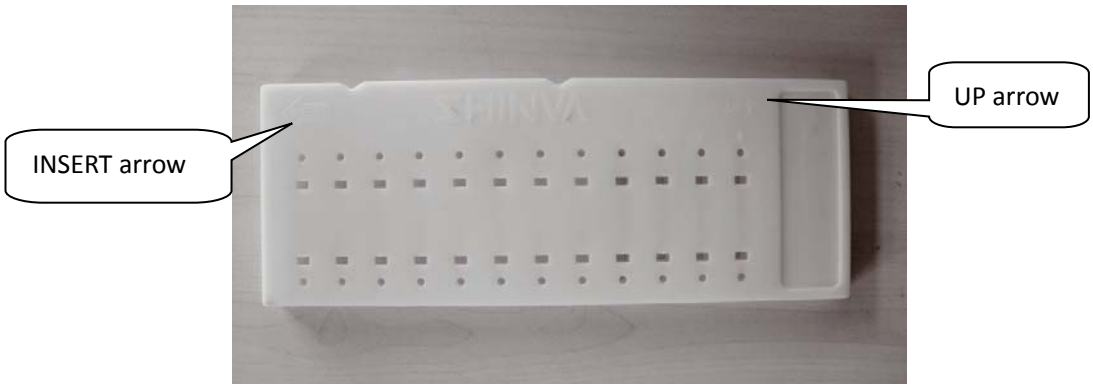




2) Remove the empty cassette. The screen prompts '**Please insert new cassette!**'

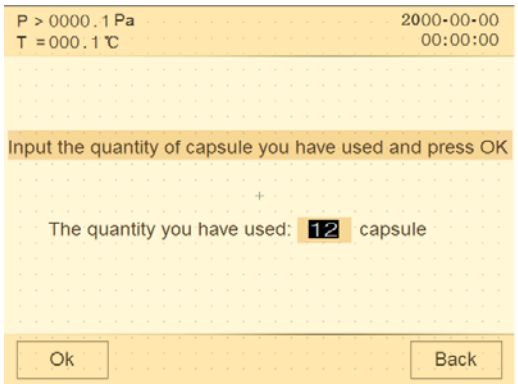


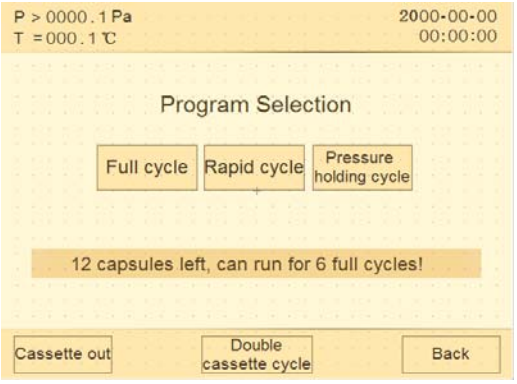
3) Insert a new cassette in correct way according to the direction of arrows imprinted on the cassette (see the picture below). Push the cassette forward till it cannot move further.



H_2O_2 cassette

4) The system prompts '**Input the quantity of capsule you have used and press OK!**' If the cassette you have inserted is a new one please push the OK button directly. Otherwise, input the number of capsule you have used and press OK button.

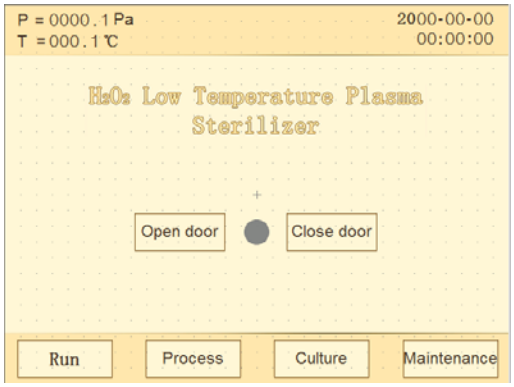




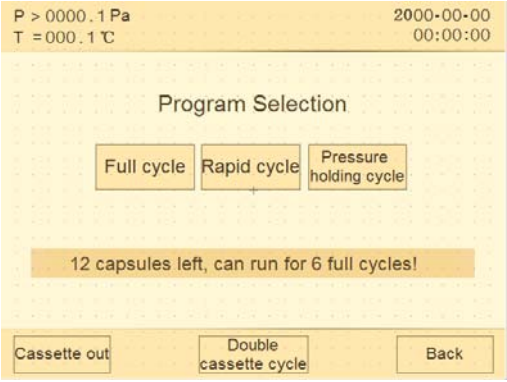
The cassette should be gently inserted into the slot in vertical direction.

5.3.2 Withdraw the cassette

1) Push the '**Run**' button on the 'Main menu' to enter into the Program Selection interface.

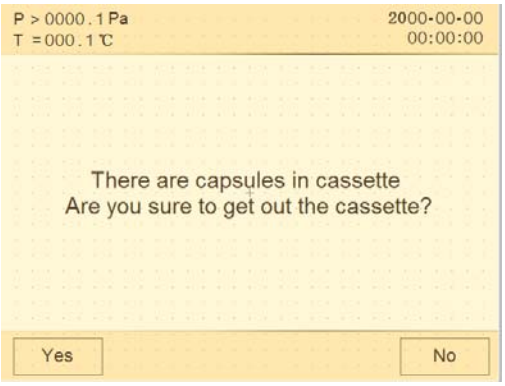


Main menu

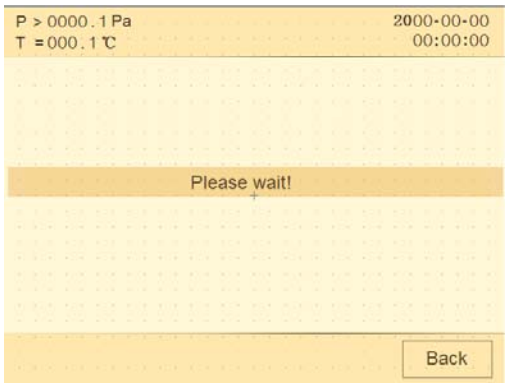


Program selection interface

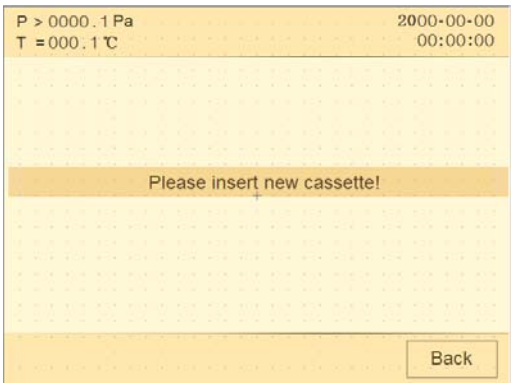
2) Press the '**Cassette Out**' button. If there are residual capsules in cassette, the screen will prompt the following interface.



3) Press **'YES '**to withdraw the cassette.



The cassette is being withdrawn



The cassette has been withdrawn

5.4 Replace the printing paper

The size of printing paper is 57 (width) mm x 30 (diameter) mm.

a. Open the printer lid and remove the paper roll.

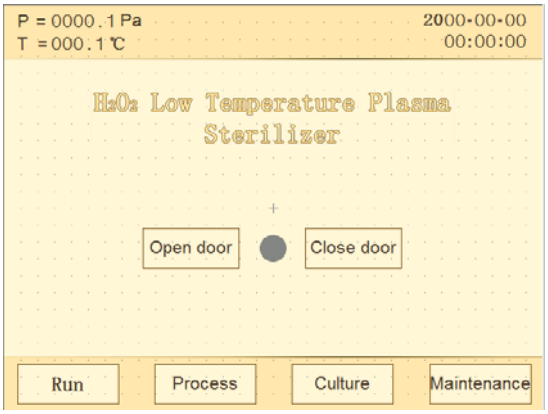
Lift here to open the lid.

b. Install the new paper roll in correct way (see the picture below) and shut the lid.

5.5 Operation menu

5.5.1 Starting up

Turn on the power supply and push the ON button on the front operation side to activate the sterilizer.



Main menu

The current pressure value and system time are displayed on the top left and right of the main menu respectively. Button description:

Run---to enter into program selection interface;

Process---to view the sterilization process;

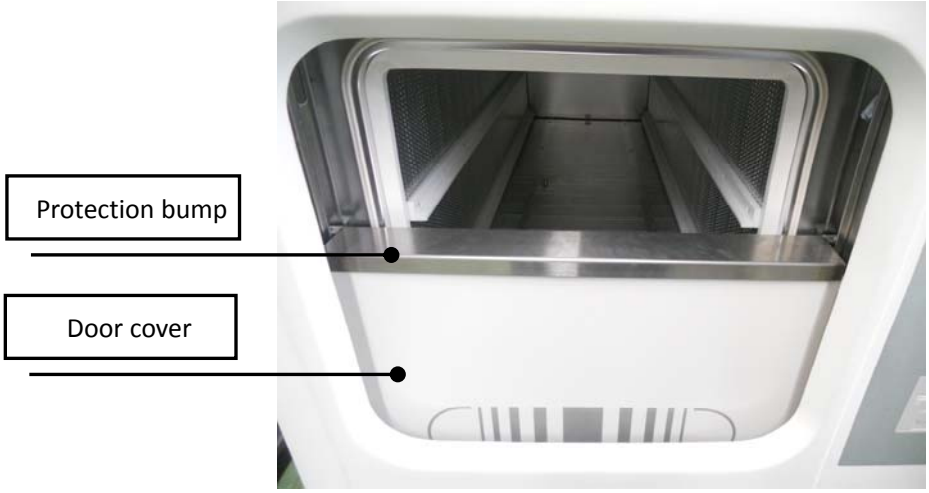
Culture---to print the biological culturing result;

Maintenance---to set up system parameters;

Open door/Close door---to control the chamber door.

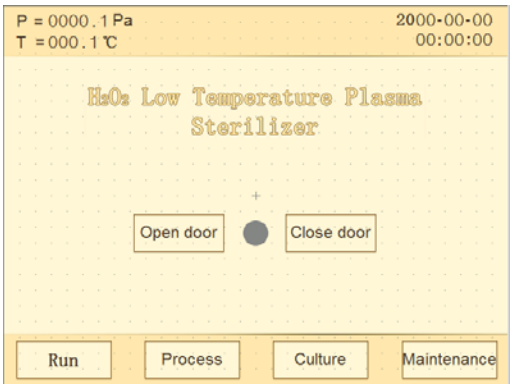
The chamber door can be controlled by 3 ways: kicking foot switch; push the door button on the front operation side; push the 'Close door' or 'Open door' button on the screen.

Before closing the chamber door, please make sure the door track is clear and free of any foreign matters. The protection bumper is installed in the chamber door as protective device to prevent operator or sterilizing articles from crush injury. Simply push the protection bumper in any emergent situation to stop the moving door. The protection bumper will not act when the door is opening.

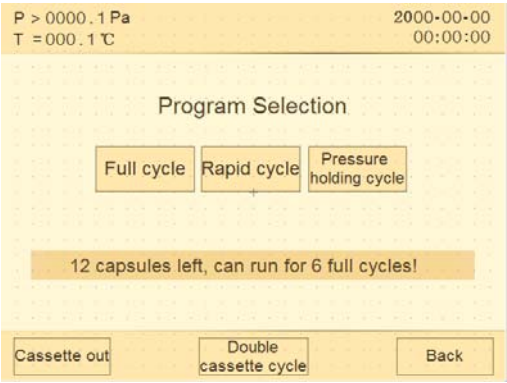


5.5.2 Program operation

Push the '**Run**' button on the main menu. System enters into the program selection interface. There are 3 cycles available to be selected.



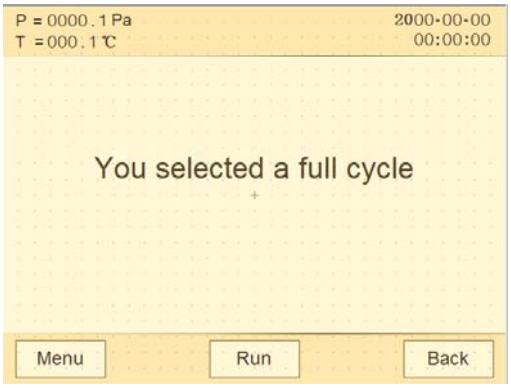
Main menu



Program selection interface

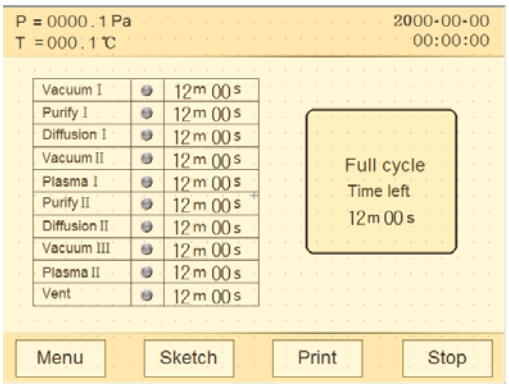
1) Full cycle and Rapid cycle

In program selection interface, push the '**Full cycle**' to enter into the following interface:



Full cycle interface

Make sure the instruments that to be sterilized and testing indicators are properly put in place; the chamber door is closed and chamber has been heated up. Push the '**Run**' button to enter into the following interface:

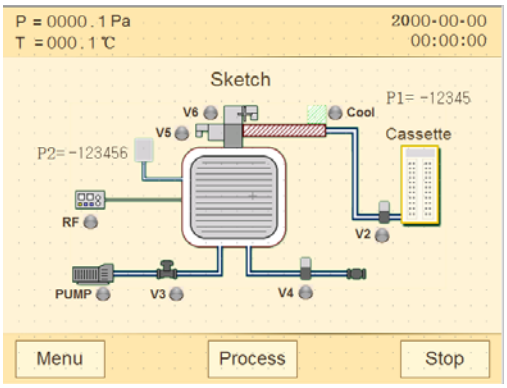


Running interface

During the program running, there is a yellow indicating light on each of the interface. This light will be off once the current program is done. The display of current stage changes from white to yellow and the corresponding timer starts working while the program runs to a certain stage. The display changes back to white and the timer stops working, once the current stage finishes. The main timer shows the running time of the entire sterilization program.

Push the '**Menu**' button to go back to the main menu.

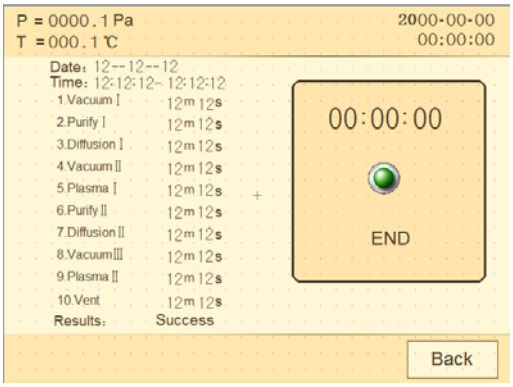
Push the '**Sketch**' button to see the following sketch interface.



Sketch interface

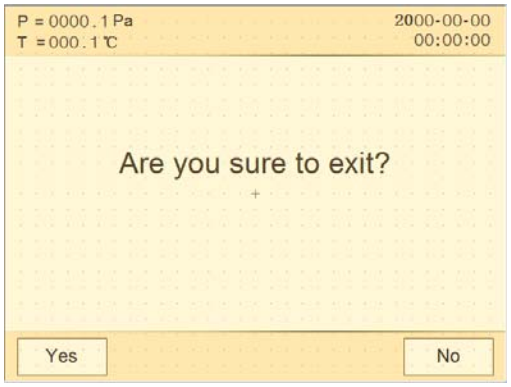
The Sketch interface shows the sterilization principle in a transparent way. The light indicates the component working status. For instance, while the '**PUMP**' is working, the light on the interface that indicates '**PUMP**' will on.

Once program is finished the screen automatically switches to the sterilization report interface, while the buzzer sounds for 10 seconds. Push the '**Back**' button to go back to the main menu.



Sterilization report interface

Push the **'Stop'** button to terminate the program when there is an emergency or alarm. You will see the following interface:



Reset prompting interface

Push '**Yes**' to terminate the ongoing program. The system will switch to vacuum-breaking status till the pressure in chamber back to normal. Once program is finished the screen automatically switches to the sterilization report interface, while the buzzer sounds for 10 seconds. Push the

'**Back**' button to go back to the main menu.

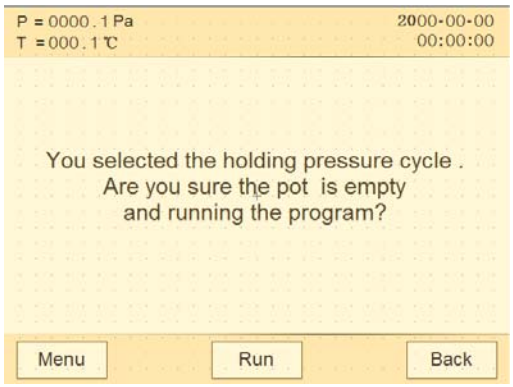
If there is an unexpected accident happens the system will automatically display alarm screen with a one-second intermittent sound.

If there is an overtime alarm during the sterilization, we recommend that push 'OK' to exit. Do not restart the sterilization until the fault is well handled.

2) Pressure holding cycle

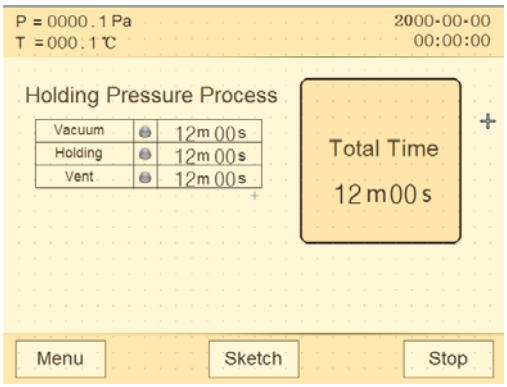
The '**Pressure holding cycle**' is designed for testing the performance of pressure holding. User could push this button for testing use when there is alarm.

In program selection interface, push the '**Pressure holding cycle**' to enter into the following interface:



Pressure holding cycle interface

Under the condition of that the pressure-holding premise is met (i.e. the chamber door is closed and the chamber has been preheated.), push the 'Start' button to activate it. Please notice that this cycle could only be operated when the chamber is empty. The following interface shows the pressure-holding process:



Holding Pressure process interface

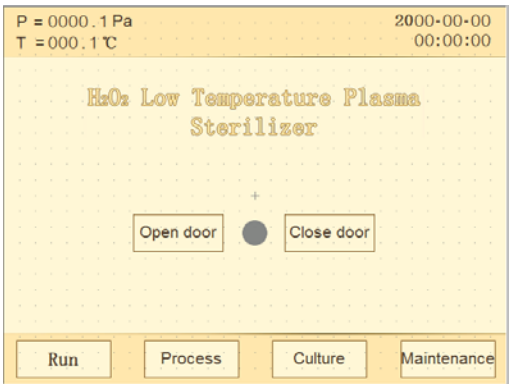
A qualified pressure-holding indicates that both the equipment and vacuum pump are working well. Otherwise, if an unqualified pressure-holding or alarm happens, it is necessary to check the equipment or contact us.

The pressure-holding cycle could only be operated when the chamber is empty.

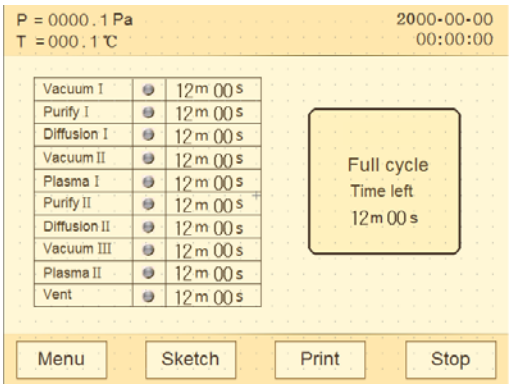
5.5.3 View and print sterilization information

a) The system will automatically record and print the last sterilization report once the sterilization cycle is finished.

For another printing of sterilization report (not applicable for standard configuration unless the gateway storage is equipped), push the '**Process**' on the main menu to enter into the process interface.



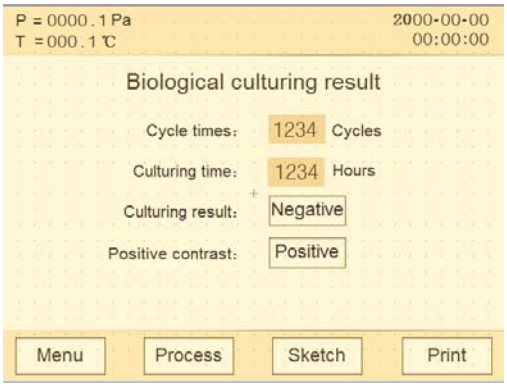
Main menu



Process interface

Push the '**Print**' button to print the report. Only the last sterilization record can be stored and will be automatically cleared by the new record.

b) To print the bio-incubation record, please push the '**Culture**' button on the main menu to enter into the biological culturing interface.



Biological culturing interface

Input the cycle times and culturing times and select the culturing result and positive contrast before pushing the 'Print' button. System will print the result automatically.

c) The previous sterilization information can be viewed and printed via the mass-memory device (only if user selected this memory device). The sterilizer will automatically save each sterilization record into the USB flash disk (it can store million times records) which attached with the memory device, and print out the record via SHINVA Sterilization View software (see the following picture) to make sure the record can be kept for long term. We will install this software into your PC.

5.5.4 Shutting down

Push the OFF button on the control panel to shut down the machine. For the heat preservation purse, please make sure the chamber door is closed when the machine is not in use.

6 Operation regulations

1). Before sterilization, the equipment should be turned on 30 minutes in advance for preheating till the displayed temperature exceeds 45℃.

2). The articles to be sterilized should meet the regulations noted in the 'Sterilization Range' section of the user manual. Operations should be carried out in strict accordance with regulations in 'Article Treatment before sterilization'.

3). Please follow the following steps for the entire operation:

Push the ON button--> preheat for 30minutes --> open the door--> put articles in chamber--> start the program--> program completed and data printing--> open the door--> take articles our from chamber--> close the door--> push the OFF button

4). The basket should be cleaned with a clean dry cloth after the final sterilization is completed every day. (Please refer to chapter 7 for details.)

5). Notes:

- Please change the H_2O_2 cassette in time when the screen prompts the cassette is running out of capsules.
- When the alarm 'vacuum overtime' occurs, please check whether the articles to be sterilized are dry. If not, make them completely dry and carry out sterilization once again. If the same alarm still occurs, please contact with us.
- When other alarms occur, please exit the program and check the equipment according to chapter 8. If the same alarm still occurs after restarting the equipment please contact with us.
- Please change the vacuum pump oil in time when the screen prompts so.
- Please contact us promptly if you have any problems.
- Please strictly follow the relevant regulations mentioned in this manual during all operations.

7 Routine maintenance

7.1 Cleaning of the sterilizer

- Please turn off the power supply before cleaning.
- Wipe the surface of the equipment with clean and dry cloth, do not use strong detergents.
- Avoid water or cleaning solution permeating into the sterilizing chamber and the touch screen.

7.2 Maintenance of the vacuum pump

- Check the transparency of oil before startup each week, and change oil if it is found that the oil is in dark color or muddy. Check the oil level before startup each week to ensure that the oil level during the pump running should be in the middle position of the oil-level window. The startup oil level should be in the 4/5 position of the oil level window. It is necessary to change oil for the vacuum pump after it runs for 100 hours for the first use. Afterwards, change the oil at least every 300 hours. After the sterilizer reaches the time for changing oil, there will be an oil change prompt on the screen. Please refer to the attached pump description or contact us for methods of oil change.
- Check and clean the dust particle filtering net every 3 months and replace it if necessary.
- Clean the built-in oil-stain filter semi-annually and replace it if necessary.

7.3 Maintenance of oil-stain filter

- Replace the filter element of the oil-stain absorption filter after each 300 operations.
- You can replace the filter element earlier if the filter element of the oil-stain absorption filter is found to be damaged, or if the smell of oil in the room is obvious.

7.4 Maintenance of H2O2 filter

- Replace the filter element of the H_2O_2 filter in every 500 times running.
- You can replace the filter element of the filter earlier if there is any oil smell.
- Contact us if you have any problems during routine maintenance.

Failure to comply with these principles may damage the sterilizer, and even make the warranties invalid if serious.

8 Fault alarm and solutions

If the faults still cannot be removed by the following solutions please contact us.

8.1 Faults and solutions

Faults	Reasons	Solutions
The screen does not work.	● The screen is not switched on. ● There is no 24V power supply. ● Part of the AC connection is abnormal.	● Check whether the power supply of the screen is reliably connected. ● Check and turn on the 24V power supply. ● Check whether the AC connection is normal; make sure the connectors are not loosened.
No response of screen after being clicked	● Abnormal connection of the communication wire between the screen and PLC. ● The operating mode of the PLC is selected by shifting the switch in STOP position. ● The PLC is faulted and the SF fault indicator light is on. ● The screen is faulted.	● Check the communication wire. ● Select the operating mode of the PLC by shifting the switch in RUN position. ● Replace a new PLC. ● Replace the screen
Printer does not work	● No printing paper in place. ● The printer paper is reversely placed. ● The power of printer is not switched on. ● The communication wire of the printer is faulted. ● The printer is faulted. ● The data memory gateway is faulted.	● Install a new roll of printer paper. ● Correctly place the printer paper. ● Check and switch on the power of the printer. ● Check whether the communication wire of the printer is reliably connected. ● Manually test the printing function. ● Check the data memory gateway.

Faults	Reasons	Solutions
Vacuum pump does not work	● The phase sequence relay protects. ● The power line of the vacuum pump is not connected. ● The circuit breaker of the pump is not closed or the contactor is not attached. ● The thermal relay of the vacuum pump protects. ● The vacuum pump is burnt out or seized.	● Change the phase sequence of any 2 phases of fire wires. ● Check whether the power line is firmly connected. ● Check the status of the pump circuit breaker and the contactor line. ● Check whether the set value of the thermal relay is proper or whether the thermal relay is damaged. ● Replace the vacuum pump.
Parameters restore to defaults and system time changes to 0	● The battery of the PLC is exhausted.	● Replace the battery of the PLC.

8.2 Alarms and solutions

Messages	Reasons	Solutions
The tank temperature T1 is wrong !	The chamber temperature is higher than set value.	● Check whether the detection of the temperature sensor is normal. ● Check whether the output point Q2.1 of the PLC works well. ● Check whether the solid-state relay is normal.
	The chamber temperature is lower than set value.	● If the equipment is just turned on, please wait for 50mins since the sterilizer is not totally preheated. ● Check whether the output point Q2.1 of the PLC and the solid-state relay work well. ● Check whether the detection of the temperature sensor is normal. ● Check whether the heating element and its wiring connection are normal.
The door temperature T2 is wrong !	The door temperature is higher than set value.	● Check whether there is output from Q2.3 (front door) and Q4.2 (back door) of the PLC. ● Check whether the relay J16 (front door) and J19 (back door) are normal.
	The chamber door temperature is lower than set value.	● If the equipment is just turned on, please wait for 50mins since the sterilizer is not totally preheated. ● If the equipment is turned on for more than 50mins, check whether there is output from Q2.3 (front door) and Q4.2 (back door) and the relay J16 (front door) and J19 (back door) are normal. ● Check whether the detection of the temperature sensor is normal. ● Check whether the heating element and its

Messages	Reasons	Solutions
		wiring connection are normal.
The steam temperature T3 is wrong !	The gasifier temperature is higher than set value.	● The heating control relay of the gasifier cannot be disconnected.
The steam temperature T3 is wrong !	The gasifier temperature is lower than set value.	● Check whether the detection of the temperature sensor is normal. ● Check whether the heating element and its wiring connection are normal.
The purifier is too cool !	The purifier temperature is lower than set value.	● Check whether the heating element and its wiring connection are normal. ● Check whether the detection of the temperature sensor is normal.
The purifier is too heat !	The purifier temperature is higher than set value.	● The heating control relay of the heating element is broken. ● The purification fan is faulted. ● Check whether the detection of the temperature sensor is normal.
Vacuum I overtime vacuum II overtime Vacuum III overtime	Vacuum overtime alarm	● Check whether the articles that to be sterilized are completely dry. ● Check whether the pressure-holding is good. ● Check whether the vacuum pump works well. ● Check whether the V3 valve is completely turned on. ● Check whether the detection of pressure transmitter P is normal. ● Check whether the valve V2 is normal.
Diffusion Ⅰ pressure is high! Diffusion Ⅱ pressure is high!	The H_2O_2 diffusion pressure is over high in diffusion cycle.	● Check whether the articles that to be sterilized contain absorbable matters. ● Check whether the solenoid valve V2 is normal. ● Check whether the diffusion valve V6 is normal. ● Check whether the vacuum valve V3 is not closed completely.
Diffusion Ⅰ pressure is low! Diffusion Ⅱ pressure is low!	The H_2O_2 diffusion pressure is over low in diffusion cycle.	● Check whether the purification solenoid valve V5 is normal. ● Check whether the V2 valve is normal.
The door is opening !	The door is not fully closed.	● Close the door. ● The travel switch of the door-close position is faulted.
The pump overheating protection !	Vacuum pump contactor overheating protection	● Check the operating conditions of the electrical appliances in the control part of vacuum pump.